Chiyo (Food Is Medicine) *Lab*

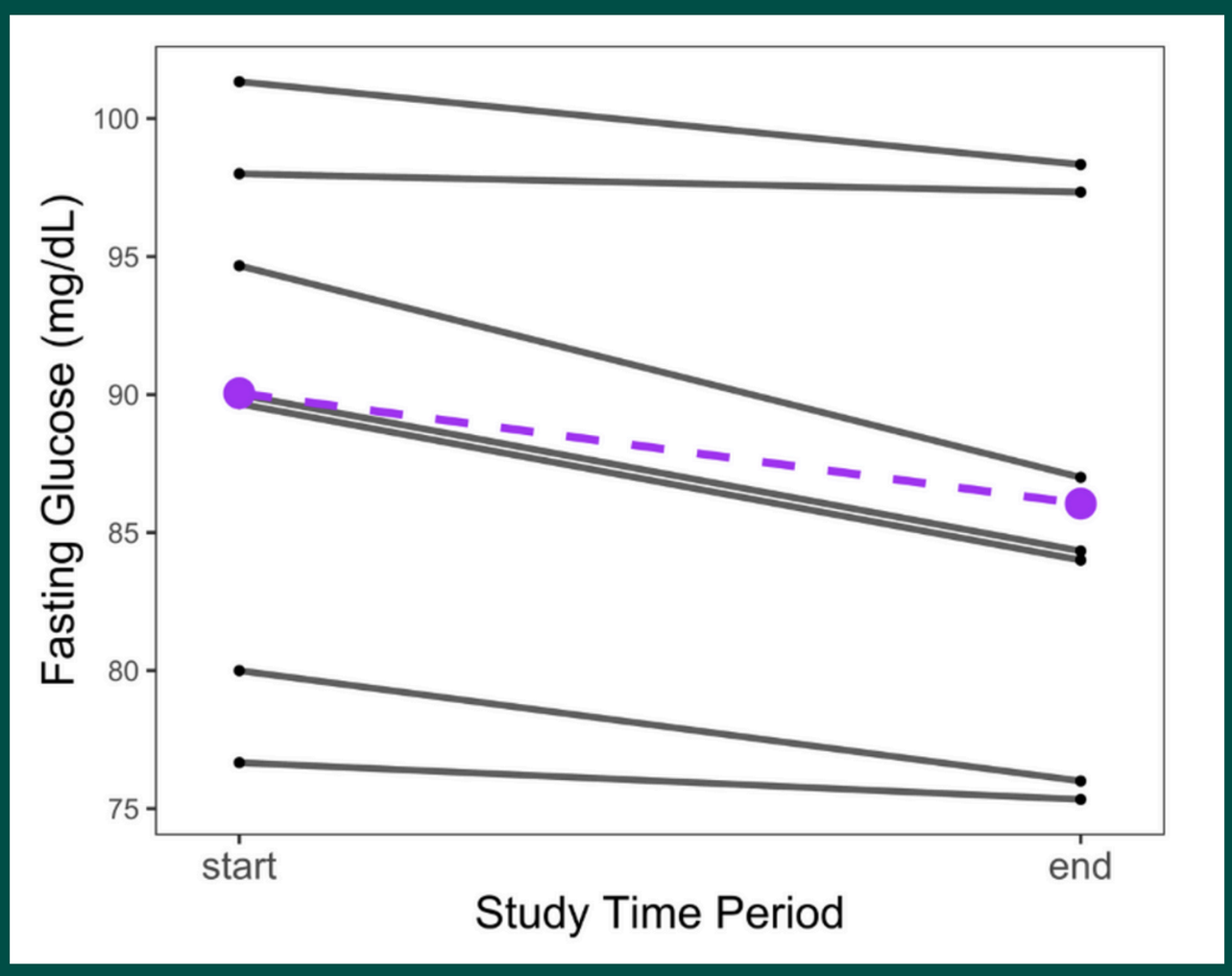
GD PILOT STUDY FINDINGS

In addition to using ingredients backed by science, Chiyo conducted a research study to evaluate the impact of our new GD Prenatal Program on glucose management in a real-life context. Participants in the study received meals from the GD Prenatal Nutrition Program and completed daily surveys logging their glucose levels before and after each meal, their physical activity, and their fluid intake. Study participants were also encouraged to make an appointment with Chiyo's registered dietitian, who helped personalize the program to their unique situation.

OVERALL RESULTS

Start of the Program Compared to the End of the Program[1]

Compared to when they started Chiyo's GD program, after an average of 4.76 weeks in the program (range = 2.5-9 weeks):



- Fasting glucose decreased from an average of 90 mg/dL to an average of 86 mg/dL, a change that was statistically significant (p = .0057) – see graph below
- Amount of exercise, hydration, and all other glucose metrics did not significantly change*

*p-values > .05 for paired t-tests, indicating that any changes were not statistically significant

1 Results were very similar when we only included participants who were not taking medication to control their GD (7 out of 9), and when we only included participants who completed the full 6-week program (4 out of 9). Chiyo's program is not designed to support individuals on medication, but we do not recommend any specific approach to management over another. Participants were allowed to participate in the study if they wanted to use Chiyo to manage their GD alongside medication, as long as they were under the care of their ob/gyn.

Graph Note: Each black line represents a participant, and the purple dotted line represents the study average. 7 out of 9 participants had enough data to be included in this analysis.

Exercise and Hydration

During the program, participants reported...

- An average of 34 minutes of physical activity per day, above the recommended amount of 30 minutes per day for GD management
- Outside of broths and tonics included in Chiyo's GD Program (16 oz per day), participants reported drinking an average of 82 oz of fluid per day, above the minimum amount (60 oz) and approaching the ideal amount (84 oz) recommended for GD.

Glucose Management

During the program, all blood glucose metric averages (mg/dL) were in the ideal range for GD management (1):

Fasting Glucose	24-hour Glucose	Pre-Meal Glucose	1-hour Post Meal Glucose	2-hours Post Meal Glucose	Post Meal Blood Sugar Elevation
88 mg/dL	105 mg/dL	97 mg/dL	121 mg/dL	108 mg/dL	26 mg/dL
<90 mg/dL ideal	*<115 mg/dL ideal*	*<105 mg/dL ideal*	*<140 mg/dL ideal*	*<120 mg/dL ideal*	*<30 mg/dL ideal*

Who Participated in The Study?

Sample Size = 9

Participants were between 30 and 36 weeks into pregnancy when they started the Chiyo GD program (average = 33 weeks).

Context Notes

- Insulin resistance typically increases across pregnancy (2), so if participants hadn't changed their diet at all, we would actually expect to see all glucose metrics increase across this period of pregnancy.
- Exercise becomes more difficult as delivery date approaches in the third trimester due to more pregnancy-related symptoms (such as fatigue or pain), so pregnant people generally become less active across this period (3).

So, the fact that we either saw no change in glucose metrics across the study, or even an decrease in one metric (lower fasting glucose), suggests that the program substantially supported participants in managing their GD.